

Upamanyu Ghosh

☎ +91 8210702924 — ✉ upamanyu24300@iiitnr.edu.in — 🔗 linkedin.com/in/upamanyu-ghosh-335827220/ — 📁 github.com/Upamanyu24300

Summary — Detail-oriented developer and M.Tech student specialising in AI and Data Science, with hands-on experience in building machine learning models, data pipelines, and AI-powered applications. Comfortable working across the full data lifecycle — from collection and preprocessing to model development and result communication.

Skills

Machine Learning TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Statistical Modelling
GenAI & NLP LangChain, OpenAI, Hugging Face Transformers, Chroma, FAISS, RAG, Groq, Ollama, Streamlit
Full-Stack Development MongoDB, React, Express, Node.js, Next.js, Tailwind, Vercel, Flask
Cloud & DevOps AWS EC2, S3, Lambda, Cloudflare, Linux, Docker, System Design
Languages Python, SQL, C/C++, JavaScript, TypeScript

Education

Masters in Technology IIIT Naya Raipur <i>Specialization: Artificial Intelligence and Data Science</i>	August 2024 – Present GPA – 9.0
Integrated Masters in Science (Physics) Birla Institute of Technology, Mesra	July 2017 – May 2022 GPA – 7.63

Projects

IIIT-NR Educational Chatbot – Designed an API-driven AI chatbot to answer educational queries related to the IIIT-Naya Raipur website using Retrieval-Augmented Generation (RAG). – Built a modular data ingestion pipeline incorporating web scraping, OCR, and document processing to extract and clean text from institutional sources. – Implemented a vector-based retrieval system using FAISS/Chroma for context-aware responses, with logging and fallback handling for robustness.	Ongoing
Next Word Prediction – Developed an LSTM-based next-word prediction model trained on Shakespeare’s <i>Hamlet</i> , applying NLP preprocessing pipelines including tokenisation and sequence padding using TensorFlow and Keras. – Analysed model performance across training epochs using Matplotlib; tracked loss and accuracy metrics to evaluate generalisation. – Deployed the model through an interactive Streamlit interface, enabling real-time inference and demonstrating results to non-technical audiences.	Jan 2025 – Mar 2025
Visualization of Algorithms – Developing a web-based platform to explain data structures and algorithms through interactive visualisations and AI-assisted code explanations. – Built a backend inference service that generates natural language explanations of code logic using LLMs, consumed by a frontend visualisation interface. – Implemented step-by-step algorithm execution with interactive playback controls; added input validation and request controls for safe usage. – Project Link: visualization-of-algorithms.vercel.app	Ongoing

Experience and Achievements

Build91 Software Developer Intern – Developing AI-driven video tools focusing on generating consistent videos using SORA and multiple AI agents. – Contributing to AI-powered interior design solutions and Buteak Hotels’ website and CRM development. – Performed full-stack development using Node.js and Python, with cloud deployment on AWS EC2 instances. – Deployed Python-based chatbots and RAG systems on AWS EC2 (t3.small) for scalable AI solutions. – Designed and exposed REST APIs hosted on EC2, enabling automation workflows and integration with external platforms such as Make.com and WhatsApp. – Created technical documentation and deployment guides to support system maintenance and future scalability.	Oct 2025 – Present Raipur, Chhattisgarh
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– AIR 2729 | 96.39 percentile | GATE Score – 513

IBM Qiskit Fall Fest

Sept 2023

Organizer

- Led planning and execution of the IBM Qiskit Fall Fest, coordinating workshops and technical events.
- Collaborated with IBM Qiskit representatives to facilitate expert-led sessions on quantum computing.

Government of India

May 2019 – July 2019

District Development Associate

- Contributed to the *Wonder On Wheels* (Mobile Science Laboratory) initiative; prepared study materials and operational manuals for on-site science experiments.